Assignment No.2 2413FEB2T1 : Engineering Mathematics-2
Topic: Linear Differential Equations with Constant Coefficients

Problems on Associated Equation

1. Solve $\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 6y = 0.$
2. Solve $\frac{d^4y}{dx^4} + 2\frac{d^2y}{dx^2} + y = 0.$
3. Solve $(D^4 - 2D^2 + 1)y = 0.$

Problems on $X = e^{ax}$

1. Solve $(D^2 + D - 6)y = e^{4x}.$
2. Solve $(D^2 + D)y = e^{4x}$
3. Solve $(D^2 + D - 6)y = sinh3x$
4. Solve $\frac{d^3y}{dx^3} + 4\frac{dy}{dx} = cosh^3x$

Problems on $X = sinax$ or $X = cosax$

1. Solve $(D^4 - a^4)y = sinax$
2. Solve $(D^3 + D^2 + D + 1)y = sin^2x$
3. Solve $(D^2 + 4)y = cos2x$
4. Solve $(D^2 - 4)y = cos^2x$
5. Solve $(D^2 + 5D + 6)y = sin^2x.cosx$

Problems on $X = x^m$

1. Solve $\frac{d^3y}{dx^3} - 2\frac{dy}{dx} + 4y = 3x^2 + 5x - 3$
2. Solve $(D^3 - D^2 - 6D)y = (1 + x^2)$
3. Solve $(D^2 + 2D + 5)y = (1 + x^2)$
4. Solve $(D^3 - D^2 - 6D)y = x^2 + 1$

Problems on general formulae

1. Solve $(D^2 + a^2)y = secax$
2. Solve $(D^2 - 1)y = e^{-x}sin(e^{-x}) + cos(e^{-x})$
3. Solve $(D^2 - D - 2)y = 2logx + \frac{1}{x} + \frac{1}{x^2}$

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Problems on using method of variation of parameters

1. Solve $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = \frac{e^{3x}}{x^2}$ using method of variation of parameters.
2. Solve $(D^2 - 2D - 2)y = e^x tanx$ by method of variation of parameters.
3. Solve $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = e^{e^x}$ using method of variation of parameters.
4. Solve $(D^2 - 1)y = \frac{2}{1 + e^x}$ by method of variation of parameters.

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